ARTICLE

Financing the Foundation: How India's New Transition Bond Framework Unlocks Industrial Decarbonization

This article throws light on the framework of IFSCA for Transition Bonds. This framework is expected to function as the architect's blueprint, by mandating credible transition plans, taxonomic alignment, and transparent disclosure mechanisms that enables foreign investors to fund the modernization of India's hard-to-abate sectors. This regulatory framework acknowledges a fundamental truth: achieving economy-wide decarbonization requires not just building the new, but systematically upgrading what already exists and cannot be abandoned.

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INTRODUCTION

Imagine attempting to renovate a centuries-old mansion while its residents continue their daily lives within its walls. This captures the predicament facing hard-to-abate sectors in their decarbonization journey. Unlike constructing an entirely new green building on vacant land, which parallels how GSS+ bonds have, successfully primarily financed renewable energy and electric vehicle infrastructure, the transformation of steel, cement, chemicals, aviation, and shipping requires retrofitting essential systems while maintaining uninterrupted operations. As these sectors form the structural foundations of modern economies, producing materials and services that will remain indispensable even in a post-zero world and hence renovation is precisely what net-zero demands. Collectively, these sectors account for roughly two-fifths of global GHG emissions[1], these industries must undergo comprehensive transformation.

Just as restoring an old mansion requires both vision and precision, the transition of heavy industries demands financing models that can support gradual, credible transformation. Traditional green finance has excelled at funding new, low-emission projects such as solar farms or electric mobility. However, these instruments often fall short when it comes to decarbonizing legacy industries that cannot simply be rebuilt from scratch. This is where transition finance steps in, bridging the gap between today's carbon-intensive reality and tomorrow's low-carbon economy. For India, whose industrial backbone continues to power both domestic growth and global supply chains, developing such financing mechanisms is not just an environmental imperative but an economic necessity.

The IFSCA's Framework for Transition Bonds intends to function as the architect's blueprint for this complex renovation, by mandating credible transition plans, taxonomic alignment, and transparent disclosure mechanisms that enables foreign investors to fund the modernization of India's hard-to-abate sectors. This regulatory framework acknowledges a fundamental truth: achieving economy-wide decarbonization requires not just building the new, but systematically upgrading what already exists and cannot be abandoned.

THE GLOBAL LANDSCAPE OF TRANSITION FINANCE

1. **Global Disclosure Developments**

   Over the past few years, the global architecture for transition finance has evolved from voluntary principles into structured, mandatory systems. The most significant development is the establishment of a global baseline for disclosure, led by the International Sustainability Standards Board (ISSB). The IFRS S2

1. *https://www.weforum.org/publications/net-zero-industry-tracker-2024/*

standard, strengthened by new 2025 guidance[2], now requires companies to disclose transition plans if they materially affect climate risk understanding. This global baseline is being implemented across major jurisdictions. The European Union (EU) through its Corporate Sustainability Reporting Directive (CSRD)[3], will see its first disclosures in 2025, while the United Kingdom's (UK) Transition Plan Taskforce (TPT) framework[4] is expected to embed into regulatory requirements by the Financial Conduct Authority (FCA). These developments mark a decisive shift, from optional sustainability reporting to legally enforceable transition accountability.

2. **Taxonomy Evolution and Market Alignment**

Parallel to disclosure reforms, taxonomies defining "green" and "transition" activities have advanced rapidly. These frameworks give investors a common language to assess the credibility of decarbonization efforts.

- Singapore has emerged as a leader through its Singapore-Asia Taxonomy for Sustainable Finance[5], introducing an "amber" category for credible transition activities. This taxonomy supported its first $500 million transition bond in October 2025.
- The ASEAN Taxonomy for Sustainable Finance[6] and the Australian Sustainable Finance Taxonomy[7] similarly integrate transition activities as distinct classifications.
- The EU Platform on Sustainable Finance, in January 2025, issued detailed guidance[8] for evaluating high-quality corporate transition plans, linking disclosure standards with activity classification.

Beyond governments, global coalitions such as the G20 Sustainable Finance Working Group, OECD, and ICMA are shaping common standards. In November 2025, ICMA launched the Climate Transition Bond Guidelines (CTBG)[9], a dedicated instrument-level framework that complements entity-level practices under the Climate Transition Finance Handbook (CTFH).

Together, these developments underscore a growing consensus: transition finance is no longer peripherality, it is becoming a core pillar of sustainable finance, grounded in disclosure, taxonomy, and transparency.

INDIA'S CLIMATE AMBITION AND THE TRANSITION FINANCE IMPERATIVE

At COP26 in Glasgow (November 2021), Prime Minister Narendra Modi announced India's net-zero emissions target by 2070 and updated the country's Nationally Determined Contributions (NDCs). India's climate finance requirements to achieve those targets vary significantly depending on scope, sectoral coverage, and time horizon, as demonstrated in the table below:

Source	Year	Estimates to 2030	Estimates to 2050/2070	Remarks
CEEW-CEF[i]	2021	Part of USD 10.1 trillion.	USD 10.1 trillion to 2070 (net-zero).	Investment gap: USD 3.5 trillion, Concessional finance need: USD 1.4 trillion.
Climate Policy Initiative[ii]	2024	USD 170 billion/year (INR 11 trillion/year).	Not specified to 2070.	Covers mitigation sectors, 87% domestic sources.
NITI Aayog India Energy Security Scenarios (IESS) 2047[iii]	2025	USD 250 billion/ year (energy transition only).	USD 250 billion/year to 2047.	Energy transition only, excludes EV infrastructure and demand sectors.
Centre for Social & Economic Progress[iv]	2025	USD 467 billion (2022-2030, four sectors namely Steel, Cement, Power and Road transport only).	Not specified.	Steel (USD 251Bn), Cement (USD 141Bn), Power (USD 57Bn), Transport (USD 18Bn), 1.3% of GDP annually.

India's climate finance landscape between 2020 and 2025 presents a sharp paradox. A raft of supportive policies has successfully "crowded in" significant capital for "green" sectors, particularly renewable energy and electric mobility. However, this success has created a profound sectoral imbalance, leaving the capital-intensive, hard-to-abate industrial sectors, such as steel, cement, chemicals, and heavy transport, facing significant barriers to accessing the transition finance required for their decarbonization.

CPI in its 2024 report[10], tracked total green finance flows of $65 billion in FY 2021-22. The sectoral distribution of this capital is heavily skewed. Clean Energy (primarily solar and wind) and Clean Transport (overwhelmingly

2. *https://www.ifrs.org/news-and-events/news/2025/06/ifrs-publishes-guidance-disclosures-transition-plans/*
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10. *https://www.climatepolicyinitiative.org/publication/landscape-of-green-finance-in-india-2024/*

electric vehicles) attracted $23.9 billion and $9 billion, respectively. Together, these two "green" sectors accounted for 53% of all tracked green finance. The said reports systematically exclude chemicals, petrochemicals, aviation, shipping, and oil & gas transition finance. This omission reflects both data scarcity and the absence of established financing frameworks for these sectors' transition pathways.

As per the Centre for Social and Economic Progress's 2025 study indicates that steel sector requires USD 251 billion (2022-2030) and Cement needs USD 141 billion (additional capital expenditure) for decarbonization efforts. This imbalance is not accidental, it is the direct result of targeted, successful policy interventions aimed at de-risking and incentivising mature green technologies. Against this, hard-to-abate sectors face a combination of policy gaps, market perception issues, and technological risk that deter competitive capital. This has resulted in a situation where investors are comfortable financing inherently "green" assets, they remain hesitant to fund "greening" activities within "brown" industrial sectors. "Framework for Transition Bonds" issued by IFSCA on July 29, 2025, aims to address policy gaps for the Indian and other issuers from hard-to-abate sectors.

IFSCA'S TRANSITION BOND FRAMEWORK: BRIDGING POLICY AND PRACTICE

> IFSCA framework has the potential to anchor GIFT-IFSC as a regional hub for transition finance, setting benchmarks for emerging economies, balancing growth and decarbonization.

1. **Purpose and Context**

   India's industrial decarbonization challenge requires a financing mechanism tailored to its high-emission sectors where transition is gradual and capital intensive. The IFSCA Framework for Transition Bonds is designed precisely to fill this policy and financing gap. By allowing companies in hard-to-abate sectors to issue bonds at GIFT IFSC while adhering to credible transition standards, the framework seeks to unlock foreign capital and build investor confidence through transparency, verification, and accountability.

2. **Structural Design and Core Principles**

   The framework is built on four core pillars that collectively strengthen its credibility and investor appeal:

   - Credible entity-level transition plans aligned with the Paris Agreement and supported by measurable emission-reduction targets.
   - Taxonomic alignment with recognized international frameworks or technology roadmaps to ensure consistency and comparability.
   - Mandatory independent external reviews such as second-party opinions, certification, or verification, to safeguard investor trust.
   - Comprehensive disclosure requirements, both at issuance and annually, covering governance, emission metrics, and capital expenditure rollout.

   These pillars directly address global investors' expectations for clarity, accountability, and science-based alignment, key prerequisites for scaling transition finance.

3. **Taxonomic Flexibility and Global Alignment**

   Recognizing that India's own Climate Finance Taxonomy remains in draft form, IFSCA has adopted a pragmatic approach. Issuers are permitted to reference internationally recognized taxonomies and technology roadmaps, such as:

   - EU, Japan, Singapore-Asia, ASEAN, and Australian frameworks, and
   - Japan's METI and the International Energy Agency (IEA) decarbonization roadmaps.

   This flexibility provides immediate functionality for Indian issuers while maintaining international credibility and minimizing transaction costs, crucial for attracting foreign institutional capital familiar with these benchmarks.

4. **Disclosure Architecture and Governance Integrity**

   Drawing on ICMA's Climate Transition Finance Handbook, the framework mandates detailed disclosures across four key dimensions:

   - Transition Plan & Governance: Measurable GHG reduction targets, board-level oversight, and linkage with overall sustainability strategy.
   - Business Model Materiality: Project-level relevance, sectoral emissions footprint, and Scope 3 emissions reporting timeline.
   - Science-based Strategy: Transparent use of modeling frameworks such as SBTi, ACT, and IEA scenarios.
   - Implementation Transparency: Capital expenditure roadmap, timeline for phasing out incompatible activities, and disclosure of locked-in emissions.

   Together, these requirements institutionalize credibility and ensure that transition bonds reflect authentic, verifiable climate action.

5. **Integration with Domestic Policy Ecosystem**

   IFSCA Framework for Transition Bonds create a powerful policy synergy with India's newly notified Greenhouse Gases Emission Intensity (GEI) Target Rules, 2025[11], operationalizing the Carbon Credit

[11] *https://beeindia.gov.in/sites/default/files/Greenhouse_Gases_Emission_Intensity_Target_Rules_2025.pdf*

Trading Scheme (CCTS) effective from fiscal year 2025-26. Hence, establishing a dual-pathway financial model for hard-to-abate sectors. This has the potential to create a triple financial value proposition for transition-focused companies as given below:

- Access to cost-effective foreign capital through transition bonds to finance low-carbon infrastructure, technology upgrades, and decarbonization pathway.
- Compliance revenue through carbon credit generation and trading when emissions reductions exceed intensity targets.
- Regulatory certainty within India's evolving carbon market framework. This complementarity addresses a fundamental challenge in hard-to-abate transition finance i.e. the extended capital payback periods (often 15-25 years for CCS infrastructure) that make pure debt financing unattractive without auxiliary revenue streams. By combining transition bond financing with carbon credit monetization under the GEI framework, India's policy architecture converts compliance obligations into financial opportunities.

6. **Outlook and Potential Impact**

The framework explicitly targets hard-to-abate sectors such as steel, cement, chemicals, aviation, shipping etc. Industries such as steel and cement alone are estimated to require USD 392 billion in investment by 2030. If the IFSCA succeeds in mobilizing even 10% of the capital requirements, taking conservative estimates, for the steel and cement sectors during 2025–2030, and if this effort is complemented by similar mobilization in other hard-to-abate sectors, it could potentially facilitate around USD 10 billion in annual investments by 2030.

STRUCTURAL BARRIERS TO TRANSITION FINANCE

In spite of the tremendous potential of transition finance, the full utilization of the IFSCA's "Framework for Transition Bonds" is facing following barriers:

a) **Lack of Climate Finance taxonomy:** India's draft Climate Finance Taxonomy (released May 2025) acknowledges the need to address *"transition, in line with country circumstances, in hard-to-abate sectors"* with Iron, Steel, and Cement as priority sectors.

b) **Sector Specific Transition Roadmap:** The Asia Investor Group on Climate Change's consultation response[12] to India's taxonomy highlights that *"the government could consider issuing guidance for corporate transition plans, aligned with national targets and sectoral roadmaps to help standardize disclosures and improve investor confidence in transition-aligned activities".*

c) **Risk Perception:** Hard-to-abate sectors face multiple risk layers such as technology risk (CCS, Green hydrogen), asset stranding risk and long payback period.

CONCLUSION: BUILDING INDIA'S TRANSITION FINANCE FUTURE

The IFSCA Framework for Transition Bonds marks a significant milestone in India's sustainable finance journey. While green finance has successfully driven growth in renewable energy and electric mobility, the next frontier lies in transforming the country's industrial core, the hard-to-abate sectors that power its economic engine. The framework fills this critical gap by introducing a credible, transparent, and globally aligned mechanism to channel capital into these sectors' decarbonization pathways.

Yet, its success will depend on implementation depth, policy convergence, and market adoption. The transition finance ecosystem in India still faces foundational challenges as enumerated in the article. The coordinated approach among regulators, industry bodies, government and financial institutions to standardize disclosures, validate credible transition plans, and provide blended finance instruments that reduce perceived risks will serve to address these challenges.

Going forward, India's transition finance architecture will evolve from being regulation-driven to market-led. The real test will be whether private capital, domestic and global, views transition finance not as compliance, but as opportunity. By linking financial performance with measurable climate outcomes, instruments such as transition bonds can reshape how capital markets support industrial competitiveness and sustainability.

IFSCA framework has the potential to anchor GIFT-IFSC as a regional hub for transition finance, setting benchmarks for emerging economies, balancing growth and decarbonization. In doing so, India would not only steer its own industrial transition but also demonstrate how developing nations can finance climate ambition without compromising development priorities.

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